River-Sharing Problem with Floods

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Abstract: In international river basins, some countries experience droughts while others face floods. Water inflow from upstream neighbors benefits drought-stricken countries but harms flood-stricken countries. This situation is not captured by existing river-sharing models (Ambec and Sprumont 2002; Ambec and Ehlers 2008). Here, we generalize these models by considering the effect of flooding. In our model of the river-sharing problem with floods, we provide two characterizations of benefit function profiles for which the downstream incremental distribution is a unique welfare distribution that satisfies both the upper bound and welfare lower bound of river basin countries. Additionally, we propose a strategic bargaining process that implements this welfare distribution as the unique subgame-perfect equilibrium.

Keywords: River sharing, Disutility of water inflow, Implementation, Pessimistic core lower bound

JEL classification: C71, C72, C78, D61, Q34

1 Introduction

The 21st century has been described as an era of 'water wars' (Postel 2006). Water wars are prominent in international rivers, which flow through at least two states. Recent climate change has resulted in droughts and floods in international rivers (Dinar et al. 2010). For example, climate change has led to frequent flooding in China, which lies along the upper reaches of the Mekong River. China has many dams in its upper river basins. Vietnam and Cambodia, located downstream of the Mekong River, often suffer from droughts caused by these dams and experience severe water shortages. Moreover, international rivers, such as the Rhine River, are fed by glaciers and snowmelt, which are the water sources of upstream countries. Countries upstream of these rivers experience water shortages as the amount of these glaciers and residual snowmelt decreases due to global warming. By contrast, countries downstream of such rivers experience flooding as snowmelt water flows in larger-than-expected volumes.

It is important to understand how countries allocate water and compensate each other along international rivers, as droughts and floods are worsening due to climate change. We aim to investigate how international doctrines and negotiations coordinate agreements regarding an international river with floods from cooperative and noncooperative game-theoretic perspectives. We extend the models developed by Ambec and Sprumont (2000, 2002) and Ambec and Ehlers (2008). These pioneering authors considered a situation in which the water inflow to each country does not decrease its welfare, assuming that each country has at most one bliss point for water consumption. By contrast, our model incorporates a water inflow threshold for each country: welfare depends only on water consumption when the inflow does not exceed the threshold, and welfare is negatively affected by the inflow when it exceeds the threshold. By specifying this threshold, our model enables the description of situations in which a country consumes water either in a drought type (Ambec and Sprumont 2000, 2002; Ambec and Ehlers 2008) or a flood type (which may be subject to the disutility of water inflow). Our model also accounts for countries that experience droughts when the inflow is scarce but experience floods when the inflow exceeds a threshold.

In practice, international law and negotiations have been used to resolve international river disputes. An international river is a common resource among watercourse countries, and property rights over water cannot be clearly defined among such countries. Thus, the following international doctrines have been developed in the context of international environmental law. *Absolute territorial sovereignty* (ATS) allows a country to assert exclusive rights to water resources within its territory without considering its impact on other countries. Specifically, this doctrine states that a basin country does not have to consider

its impact on downstream countries. *Unlimited territorial integrity* (UTI) is the opposite of ATS, stating that a downstream country can claim rights to all upstream water resources. International environmental law recommends that involved countries establish an international river commission to resolve water conflicts.[1] The amount of water to be released by each country and monetary transfer are to be determined through negotiations between commission members. For instance, the Nile River Basin Commission has helped to manage conflicts on the Nile River.[2]

To reconcile ATS and UTI, Ambec and Sprumont (2002) derived a lower bound for each coalition's welfare from ATS and an upper bound for each coalition's welfare from UTI. Under ATS, a group of countries guarantees welfare that gains on its own. This welfare is called **the core lower bound** of the coalition. The core lower bound of a coalition depends on the coalition structure of countries outside the coalition. The coalition structure of countries outside coalition S can be naturally considered to be such that S's welfare is minimized. In other words, this coalition structure means that coalition S does not face uncertainty derived from all possible actions that countries in $N \setminus S$ can take. We refer to this core as the **pessimistic core**, following the core of a partition function form game introduced by Thrall and Lucas (1963).

Our results are closely related to the following findings in the literature. From a cooperative game-theoretic perspective, Ambec and Sprumont (2002) showed a unique welfare distribution that satisfies core lower bounds and aspiration upper bounds.[3] This distribution is called the **Downstream Incremental distribution** (DI distribution), in which each country's welfare is its marginal contribution to the coalition of its upstream countries. From a noncooperative game-theoretic perspective, Ambec and Sprumont (2000) proposed a strategic bargaining process that implements only the DI distribution in subgame-perfect equilibrium. Ambec and Ehlers (2008) extended the Ambec and Sprumont (2000, 2002) model by allowing each country's welfare from its water consumption to have a satiation point. However, as water release is

[1]The following environmental laws play a significant role in the management of international rivers: the Helsinki Rules on the Uses of the Waters of International Rivers (Helsinki Rules) and the United Nations Convention on the Law of the Non-Navigational Uses of International Watercourses (United Nations Convention). The Helsinki Rules were adopted by the International Law Association in 1966, and the International Law Commission formulated the United Nations Convention in 1997. For details, see Birnie, Boyle, and Redgwell (2009).

[2]Competition for the waters of the Nile River between Egypt, Sudan, Ethiopia, and the Lake Victoria basin states has been shifting toward cooperation through the Nile River Basin Commission.

[3]The underlying motivation of these bounds is that the application of ATS and UTI can be extended to the property rights of a group of countries (i.e., a coalition) from the property rights of each country.

assumed to be costless, water inflow never reduces welfare. This corresponds to a case where all countries experience droughts.

Our model of the river-sharing problem generalizes the existing models; in particular, it can describe a situation in which water inflows lower the welfare of a country. In our **river-sharing problem with floods**, our first main result concerns cooperative game theory, and the second concerns noncooperative game theory.

First, we identify sufficient conditions for the DI distribution to be the unique welfare distribution satisfying the *pessimistic core lower bounds* and aspiration upper bounds. One sufficient condition is that each country's optimal water consumption is equal to its inflow once the inflow exceeds the threshold and entails disutility. Under this condition, called the **dam-up condition**, during flooding, it is optimal to 'dam up' by consuming all inflow, even at a cost.

The other sufficient condition is that the set of drought-stricken countries is located upstream of the set of flood-stricken countries. This condition corresponds to a situation where global warming causes upstream countries to face water shortages as glaciers and snowmelt decrease in volume. By contrast, downstream countries may face flooding as greater-than-expected glacier and snowmelt water flows from upstream.

Conversely, we show that a violation of these conditions can lead to the nonexistence of a stable agreement. Specifically, if the *dam-up condition* is violated, or if a flood-stricken country is located upstream of a drought-stricken country, we demonstrate that a counterexample can be constructed where the pessimistic core is empty.

Second, when the pessimistic core lower bounds are satisfied, we show that a simple strategic negotiation process, called **upstream sequential bargaining** in this paper, implements only the *DI distribution*: in a unique subgame-perfect equilibrium, the corresponding welfare distribution coincides with the *DI distribution*. The negotiation process, in which a country proposes a water inflow amount and a monetary transfer to the immediate upstream country, is repeated. The first proposer is the most downstream country. If any country rejects its proposal, then the previously agreed-upon proposal is nullified, and the rejecting country becomes the first proposer in the next round. In the last round, the water inflows and money transfers are implemented as agreed between the countries that accepted. On the contrary, the countries that do not form an agreement act noncooperatively. The proposed strategic bargaining process (upstream sequential bargaining) is identical to the bargaining process of Ambec and Sprumont (2000) except in the following points. First, in their process, the proposer in each round must propose water inflows and money transfers not only for itself but also for all of its upstream countries. Second, in the final round, each country whose proposal is rejected consumes all its en-

dowment and does not have to pay for water use. In the river model of Ambec and Ehlers (2008), where each country's benefit function has a satiation point, Ambec and Sprumont's (2000) process does not implement the DI distribution in subgame-perfect equilibrium. By contrast, our process can be applied to both the Ambec and Sprumont (2002) and Ambec and Ehlers (2008) models and our model.

The remainder of this paper is organized as follows. In Section 2, we survey related literature. In Section 3, we introduce river-sharing problems with floods. In particular, a generalized benefit function is introduced to capture the direct effect of water inflow on each country's benefit. In Section 4, we show two sufficient conditions, respectively, for the DI distribution to be the unique welfare distribution that satisfies the *pessimistic core lower bounds* and aspiration upper bounds. In Section 5, we introduce the *upstream sequential bargaining* process and show that it implements the DI distribution only. Finally, Section 6 presents a potential application of our model to the supply chain of technological knowledge. The Appendix contains all proofs.

2 Related literature

Axiomatic literature: We discuss the connection between our paper and related studies other than Ambec and Sprumont (2002) and Ambec and Ehlers (2008). River-sharing problems have been modeled from two main perspectives. The first focuses on how countries allocate among themselves the benefits derived from water use. The second perspective focuses on the allocation among countries of costs associated with removing pollution caused by upstream waste. Both perspectives have been extensively studied using axiomatic approaches, often grounded in cooperative game theory. Studies about the first perspective include those of van den Brink et al. (2012), Ansink and Weikard (2012), Ambec et al. (2013), and van den Brink et al. (2014); those on the second perspective include studies by Ni and Wang (2007) and Alcalde-Unzu et al. (2015).

This study proposes a model that addresses situations where inflow, which is originally a source of benefit, can impose costs on riparian countries. Our model may be seen as integrating the first and second modeling approaches. However, it differs from the second approach in the following sense. In the second modeling approach, the damage each country experiences due to water pollution (costs) is exogenously given. In our model, the costs each country incurs due to inflow are determined endogenously, depending on the optimal consumption decisions of the upstream countries.

Implementation-theoretic literature: Instead of relying on axiomatic meth-

ods, several researchers have adopted a noncooperative game-theoretic approach based on implementation theory. Since Ambec and Sprumont (2000), researchers have been studying how solutions of games defined on graphs, derived from models that allocate benefits to countries along a river, can be implemented as subgame-perfect equilibria. Examples include van den Brink et al. (2013) and Gudmundsson et al. (2019).

Ambec and Sprumont (2000) proposed a strategic bargaining process that implements only the DI distribution in subgame-perfect equilibrium. However, their model does not deal with the effects of flooding on individual countries. Without considering coalition structures, van den Brink et al. (2013) extended this framework by proposing a strategic bargaining process for zero-monotonic games defined on cycle-free graphs. Based on the order of agents along a line graph, their process implements a marginal contribution allocation corresponding to the DI distribution.[4] This bargaining process differs from that of Ambec and Sprumont (2000) and the proposed process. Gudmundsson et al. (2019) introduced a two-stage strategic bargaining process, showing that an allocation different from the DI distribution can be implemented as a subgame-perfect equilibrium.[5]

Our paper, by contrast, demonstrates the subgame-perfect implementation of the marginal contribution allocation corresponding to the DI distribution in games defined on line graphs with coalition structures. This feature distinguishes our work from the existing studies mentioned above.

[4]The process proposed by van den Brink et al. (2013) implements all marginal worth vectors in subgame-perfect equilibrium on the class of *zero-monotonic* games (Theorem 3.1 in van den Brink et al. 2013). Applied to the river model, their mechanism is as follows. The flow of negotiation rounds is similar to that in our model, with the rounds progressing sequentially upstream, starting with the first round, in which the most downstream agent is the proposer. In each round, the most downstream agent (among all agents) proposes to all its upstream agents. If all agents accept this proposal, then the proposer gets the remaining benefits; otherwise, it leaves the negotiation after obtaining the benefits from its water source. Negotiations continue in this process until all agents gain benefits.

[5]Gudmundsson et al. (2019) proposed a two-stage bargaining game structured as follows: In the first stage, each country announces the water consumption levels for all countries. In the second stage, a proposer is randomly selected, and the remaining countries decide whether to accept or reject its proposed monetary transfer. The second negotiation stage may be repeated (potentially indefinitely) until all parties reach an agreement. The selection of the proposer is determined by a recognition function that assigns bargaining power to each country. Under the mechanism outlined by Gudmundsson et al. (2019), a distribution other than the downstream incremental distribution is achieved in subgame-perfect equilibrium.

3 Model

3.1 Base setting

This section explains our proposed river-sharing problem, which is a generalization of the **river-sharing problems** of Ambec and Sprumont (2000, 2002) and Ambec and Ehlers (2008). A river-sharing problem (N, e, b) consists of a set of agents N, initial endowments $e = (e_i)_{i \in N} \in \mathbb{R}^N_+$, and benefit functions $b = (b_i)_{i \in N}$. The novelty of this study is expanding the domain of the benefit functions. Below, we explain the details.

Consider a river represented by a line divided into n segments indexed by $i = 1, \ldots, n$, where $n \geq 2$. Each agent (or country) i is located in segment i. Let $N = \{1, \ldots, n\}$ be the set of agents. An international river flows from agent 1 (the most upstream agent) to agent n (the most downstream agent). Agent j is **downstream**[6] of i if $j > i$, and agent j is **upstream** of i if $j < i$.

Let $U_i^0 = \{1, \ldots, i-1\}$ be the set of upstream agents of i, and $U_i = U_i^0 \cup \{i\}$. For notational convenience, we write for each set $S \subset N$, $U_S = U_{\max S}$, where $\max S = \max\{i : i \in S\}$.

Each agent $i \in N$ has an **endowment** of water. Agent i's *endowment* is denoted as e_i. For each $i \in N$, let $e_i \geq 0$. The river picks up water, a private good, along its course. The quantity of water increases by e_i when the river flows through agent i. Each agent $i \in N$ consumes x_i water units. Let $E_i = \sum_{j \in U_i^0}(e_j - x_j) + e_i$ be the **water inflow** reached at agent i.[7] $E_1 = e_1$. Each agent's consumption level is bounded by E_i (i.e., $x_i \in [0, E_i]$). We also denote $E_i(x) = E_i$ for the consumption vector $x \in \mathbb{R}^N$ to state explicitly that E_i depends on the upstream agents' consumption $(x_j)_{j<i}$. $E_i(x)$ is independent of the consumption profile of i and its downstream agents, i.e., $(x_j)_{j \geq i}$. Therefore, we also write $E_i(x_1, \ldots, x_{i-1}) = E_i$.

Agent i's benefit is derived from its water consumption $x_i \in \mathbb{R}_+$ and possibly the water inflow $E_i \in \mathbb{R}_+$. For each $i \in N$, let agent i's **benefit function** be $b_i \colon \mathbb{R}_+ \times \mathbb{R}_+ \to \mathbb{R}$.

We assume that b_i is continuous in both arguments, and for any $E_i > 0$, $b_i(\cdot, E_i)$ is twice continuously differentiable and strictly concave in the first argument. Given the benefit functions, for each $i \in N$, let $b_i^*(E_i) = \max_{x_i \in [0, E_i]} b_i(x_i, E_i)$ and $x_i^*(E_i) = \arg\max_{x_i \in [0, E_i]} b_i(x_i, E_i)$. Through the strict concavity of $b_i(\cdot, E_i)$, x_i^* and b_i^* are uniquely determined. Berge's maximum theorem implies that x_i^* and b_i^* are continuous at any $E_i > 0$.

Ambec and Ehlers (2008) considered a situation in which water consumption can decrease welfare, although water inflow does not decrease welfare. In

[6] We write technical terms in **bold** when they first appear and in *italic* thereafter.

[7] In the definition of water inflow, while Ambec and Ehlers (2008) exclude e_i, our model includes e_i.

this study, we allow benefit functions to be decreasing in water inflow. Formally, we assume that each benefit function has a threshold, $\hat{E}_b$, such that if the inflow is less than the threshold, the benefit is independent of the inflow, while if it exceeds the threshold, the benefit is decreasing in the inflow. Note that $\hat{E}_b$ depends on b.

Assumption 1. For each benefit function b_i, there exists $\hat{E}_{b_i} \in \mathbb{R}_+ \cup \{\infty\}$ that satisfies the following properties:

1. $b_i(x_i, E_i) = b_i(x_i, 0)$ for each x_i and $E_i \leq \hat{E}_{b_i}$.
2. $b_i(x_i, E_i)$ is differentiable with respect to E_i if $E_i > \hat{E}_{b_i}$.
3. $b_i^*(E_i)$ is strictly decreasing in E_i if $E_i > \hat{E}_{b_i}$.

To simplify the notation, we write $\hat{E}_i = \hat{E}_{b_i}$.

An interpretation of Assumption 1 is that the inflow of water causes disutility to the agent once it exceeds a certain threshold, like a flood.

It is important to note that our benefit functions include those of Ambec and Sprumont (2000, 2002), and Ambec and Ehlers (2008); if $\hat{E}_i = \infty$ for each b_i, our model is reduced to that of Ambec and Ehlers (2008).

To identify the connections and distinctions with the previous studies, we provide several examples. Let $\bar{x}_i = \arg\max_{x_i \in \mathbb{R}_+} b_i(x_i, 0)$, which is referred to as the **satiation point**. Note that by Assumption 1, $b_i(x_i, E_i) \leq b_i(\bar{x}_i, 0)$ for any x_i and E_i.

The benefit functions proposed here can represent each agent's benefit under river situations such as drought, flood, or a combination of both.[8]

- **Drought type:** If $\hat{E}_i = \infty$, then agent i never suffers from water inflow. Such agents are called *drought-type* agents in this study. This benefit function is the one that was considered in Ambec and Ehlers (2008).
- **Flood type:** Let $\hat{E}_i \in [0, e_i]$, then agent i suffers from any water inflow exceeding its endowment e_i. Such agents are called *flood-type* agents in this study. For example, $b_i(x_i, E_i) = -\max\{E_i, 2\} + (x_i + 1)^{1/2}$. For this benefit function, $\hat{E}_i = 2$, and if $E_i \geq e_i \geq 2$, $(b_i^*)'(E_i) = \frac{1}{2}(E_i + 1)^{-1/2} - 1 < 0$.
- **Mixed type:** Let $\hat{E}_i \in (e_i, \infty)$. For example, consider $b_i(x_i, E_i) = -4\max\{E_i, 5\} - (x_i - 7)^2$ with $e_i < 5$. In this case, $\hat{E}_i = 5$. Then,

[8] Agents may face different river-sharing problems in different seasons. An agent is flood type in the rainy season, while the agent is drought type in the dry season. We thank an anonymous reviewer for this insight.

$b_i^*(E_i) = -(E_i - 7)^2 - 20$ if $E_i < 5$ and $b_i^*(E_i) = -(\max\{7 - E_i, 0\})^2 - 4E_i$ if $E_i \geq 5$. Such agents are called *mixed type* agents here.[9]

We further adopt the following condition on the benefit functions:

Assumption 2. If $x_i^*(E_i) < E_i$, then there exists $\zeta \in [0, 1)$ such that $x_i^*(E_i + \epsilon) \leq x_i^*(E_i) + \zeta\epsilon$ for any $\epsilon > 0$.

We define the preferred **water outflow** of agent i, as $E_i - x_i^*(E_i)$. Assumption 2 implies that if an agent prefers not to consume all the water inflow, then additional water inflow will increase the preferred water outflow. Indeed, $E_i + \epsilon - x_i^*(E_i + \epsilon) \geq E_i + \epsilon \cdot (1 - \zeta) - x_i^*(E_i) > E_i - x_i^*(E_i)$ for each $\epsilon > 0$.

Remark 1. If x_i^* is differentiable,[10] then Assumption 2 is equivalent to $(x_i^*)'(E_i) < 1$. Through standard comparative statics, this condition is also equivalent to the inequality $-\frac{\partial^2 b_i}{\partial x_i \partial E_i} > \frac{\partial^2 b_i}{\partial x_i^2}$, which is satisfied for any $x_i < E_i$ and $E_i > \hat{E}_i$.[11] Furthermore, $-\frac{\partial^2 b_i}{\partial x_i \partial E_i} > \frac{\partial^2 b_i}{\partial x_i^2}$ is implied by the concavity of b_i.[12]

First note that the benefit functions in Ambec and Ehlers (2008) satisfy Assumption 2. To see this, let $\hat{E}_i = \infty$. Then, by Assumption 1 (1), $x^*(E_i) = \min\{\bar{x}_i, E_i\}$. If $\bar{x}_i = \infty$, $x_i^*(E_i) = E_i$ and therefore, Assumption 2 is satisfied. Suppose that $\bar{x}_i < \infty$. If $x_i^*(E_i) < E_i$, $x_i^*(E_i) = \bar{x}_i$, and therefore, $x_i^*(E_i + \epsilon) = \bar{x}_i = x_i^*(E_i) < E_i$.

We provide another example satisfying Assumption 2:

Example 1. Suppose that $b_i(x_i, E_i) = -F_i(\max\{E_i - \hat{E}_i + \tilde{x}_i - x_i, 0\}) + \bar{b}_i(x_i) - c_i(x_i)$, where $\tilde{x}_i = \arg\max_{x_i \in \mathbb{R}_+}(\bar{b}_i(x_i) - c_i(x_i)) \leq \hat{E}_i$. Here, $F_i \colon \mathbb{R}_+ \to \mathbb{R}$ represents flood damage, which is caused by water outflow $E_i - x_i$, assuming that $F_i'(0) = 0$ and $F_i'(y) > 0$ for each positive real number $y > 0$. $\bar{b}_i \colon \mathbb{R} \to \mathbb{R}$ is the net benefit from water consumption, and $c_i \colon \mathbb{R} \to \mathbb{R}$ is the abstraction cost of water. If $\hat{b}_i'' < 0$, $c_i'' > 0$, and $F_i'' > 0$, then the inequality $-\frac{\partial^2 b_i}{\partial x_i \partial E_i} > \frac{\partial^2 b_i}{\partial x_i^2}$ is satisfied as $-\frac{\partial^2 b_i}{\partial x_i \partial E_i} = -F_i''$ and $\frac{\partial^2 b_i}{\partial x_i^2} = -F_i'' + \bar{b}_i'' - c_i''$.

By contrast, the following benefit function violates Assumption 2:

[9] This example of a mixed-typed function was created according to the advice of Ricardo Martínez. We are grateful for his useful suggestion.

[10] If $x^*(E_i) < E_i$, then $x^*(E_i)$ is an interior solution and is therefore differentiable by the implicit function theorem.

[11] Assumption 2 holds in the region where $E_i \leq \hat{E}_i$ as the benefit is independent of E_i in that region.

[12] Indeed, the strict concavity of b_i implies that $\left(\frac{\partial^2 b_i}{\partial x_i \partial E_i}\right)^2 < \frac{\partial^2 b_i}{\partial x_i^2} \frac{\partial^2 b_i}{\partial E_i^2}$ and $\frac{\partial^2 b_i}{\partial x_i^2}, \frac{\partial^2 b_i}{\partial E_i^2} < 0$. Therefore, if $\left|\frac{\partial^2 b_i}{\partial E_i^2}\right| \leq \left|\frac{\partial^2 b_i}{\partial x_i^2}\right|$, then the inequality $-\frac{\partial^2 b_i}{\partial x_i \partial E_i} > \frac{\partial^2 b_i}{\partial x_i^2}$ is satisfied.

Example 2. Assume that $b_i(x_i, E_i) = -F_i(E_i - x_i) - \frac{1}{1+E_i}x_i$, where the first term represents flood damage that appears in Example 1, and the second term is the water abstraction cost. This benefit function implies that the water abstraction cost decreases as the water inflow increases. In this case, $(x_i^*)'(E_i) = 1 + \frac{1}{F_i'' \cdot (1+E_i)^2} > 1$ if $F_i'' > 0$.

However, assuming the existence of benefit functions like that in Example 2 is inconsistent with actual dam management practices, as increasing water inflow raises the risk of flooding and makes water regulation more difficult.

In the following sections, we assume that the benefit functions satisfy Assumptions 1 and 2.

Finally, in addition to water consumption and inflow, monetary transfers are available for agents in unbounded quantities to perform side payments. Agent i's utility from consuming x_i units of water, receiving a water inflow E_i, and receiving a net money transfer m_i is given in the following quasilinear form: $\pi_i \colon \mathbb{R}_+^2 \times \mathbb{R} \to \mathbb{R}$ such that $\pi(x_i, E_i, m_i) = b_i(x_i, E_i) + m_i$. We focus on river-sharing problems (N, e, b) in which $x_i^*(e_i) = e_i$ is satisfied, following Ambec and Ehlers (2008). The condition that $x_i^*(e_i) = e_i$ enables us to focus on an environment where each individual's optimal consumption does not influence other agents, although coalition formation can.

4 Cooperative game-theoretic approach

4.1 Coalitional worth, aspiration upper bounds, and core lower bounds

Following Ambec and Sprumont (2002) and Ambec and Ehlers (2008), we take a cooperative game-theoretic approach to our river-sharing problem.

An **allocation** is a vector $(x, m) = (x_1, \cdots, x_n, m_1, \cdots, m_n) \in \mathbb{R}_+^n \times \mathbb{R}^n$ satisfying the following *feasibility* constraints:

$$\sum_{i \in N} m_i \leq 0$$

$$\sum_{i=1}^{j} x_i \leq \sum_{i=1}^{j} e_i \quad \text{for each } j \in N,$$

$$x_j \geq 0 \text{ for each } j \in N.$$

An allocation (x^*, m^*) is **efficient** if and only if $\sum_{i=1}^n m_i^* = 0$ and

$$x^* = \underset{x:x \text{ is feasible}}{\arg\max} \sum_{i=1}^n b_i(x_i, E_i(x)).$$

A tuple $z \in \mathbb{R}^n$ is an allocation of (efficient) benefits if $\sum_{i \in N} z_i = \sum_{i \in N} b_i(x_i^*, E_i(x^*))$. To achieve benefit allocation z, monetary transfer is determined by $m_i^* = z_i - b_i(x_i^*, E_i(x^*))$ for each $i \in N$.

Ambec and Sprumont (2000, 2002) and Ambec and Ehlers (2008) defined **core lower bounds** and **aspiration upper bounds**, influenced by international law doctrine.

The notion of *core lower bounds* is influenced by the international law doctrine **Absolute Territorial Sovereignty** (**ATS** for short). This lower-bound property states that the welfare of each coalition should not be lower than the welfare attainable by the water the coalition controls. *Aspiration upper bounds* are influenced by the international law doctrine **Unlimited Territorial Integrity** (**UTI** for short). This upper-bound property states that the welfare of each coalition should not be higher than what it can achieve in the absence of the remaining agents.

We formally define these notions as follows: Let $\mathcal{P}(S) \equiv \{P\colon P$ is a partition of N such that $S \in P\}$ be the set of partitions[13] that has S as one of its elements. Let $v(S, P)$ be the coalitional worth of S when the partition $P \in \mathcal{P}(S)$ is formed.

Formally, for each partition P, $v(S, P)$ is defined through the following extensive form game, (N, H, B^P). Here, N is the set of players, which is identical to the set of agents in the river-sharing problem.[14] $H = \{(x_i)_{i=1}^t : 1 \leq t \leq n-1\} \cup \{\varnothing\}$ is the set of histories. h is said initial history if $h = \varnothing$. Each history specifies the action set of the agents. When $h = (x_1, \ldots, x_{i-1})$, $A_i(h) = [0, E_i(x_1, \ldots, x_{i-1})]$, and $A_j(h) = \emptyset$ for each $j \neq i$. The strategy is a function $f_i : h \to A_i(h)$. Specifically, if $h = \varnothing$, $A_1(h) = [0, e_1]$ and $A_j(h) = \emptyset$ for each $j \neq 1$. Finally, the payoff function of the agent i is denoted by B_i^P. To define the function, let $T \in P$ be a set of agents such that $i \in T$. Then, for each $x = (x_1, \ldots, x_n)$,

$$B_i^P(x) = \sum_{j \in T} b_j(x_j, E_j(x)).$$

[13] $P \subset 2^N$ is a partition of N if $\bigcup_{S \in P} S = N$, $S \cap T = \emptyset$ for each $S, T \in P$, and $T \neq \emptyset$ for each $T \in P$.

[14] Ambec and Ehlers (2008) consider the set of players to be the partition. However, as explained below, we assume that any agents in the same coalition in the partition have the same preferences. Therefore, we can deal with the partition as the set of players in our model, as in Ambec and Ehlers (2008).

Note that $B_i^P = B_j^P$ for each $i, j \in T$. Therefore, we write $B_T^P = B_i^P$ for each $i \in T$.

For each $h \in H$ such that $h = (x_1, \ldots, x_{i-1})$, let $h_j(f, h) = (x_1, \ldots, x_j)$ such that $x_j = f_j(h_{j-1}(f, h))$ for each $j \geq i$. Strategy profile $f = (f_i)_{i \in N}$ is a *subgame-perfect equilibrium* if $B_i^P(h_n(f, h)) \geq B_i^P(h_n((f_{-i}, f_i'), h))$ for each strategy of i, f_i', each $i \in N$, and each $h \in H$.

Theorem 1 of Harris (1985) implies that this extensive form game has a subgame-perfect equilibrium.[15] Now let $SPE(\mathcal{G}^P)$ be the set of the subgame-perfect equilibria of game $\mathcal{G}^P = (N, H, B^P)$. We define the coalitional worth as the optimal outcome of the subgame-perfect equilibrium in the game.[16]

$$v(S, P) = \max_{f \in SPE(\mathcal{G}^P)} B_S^P(h_n(f, \varnothing)).$$

Remark 2. Note that this extensive form game may have multiple equilibria because passing and withholding the water may leave some coalitions indifferent between passing and withholding water.[17] In Remark 1 of Ambec and Ehlers (2008), they assume that any coalition passes any water if indifference occurs. This assumption guarantees the upper *semi*-continuity of the reaction functions,[18] as any other coalition weakly prefers any water to be passed in their model. However, in our model, while passing water is preferable for some coalitions, it may harm the other coalition, as benefit functions can be decreasing in water inflow. This may lead to a situation in which a reaction function is not upper semi-continuous; consequently, no subgame-perfect equilibrium exists under such a tie-breaking rule. Instead, we consider the optimal outcome among subgame-perfect equilibria for coalition S to formalize the coalitional worth $v(S, P)$.

Given any P, $v(S, P)$ is called a **core lower bound** of S. Ambec and Ehlers (2008) introduced two notions of *core lower bounds* as follows: If $P = \{S\} \cup \bigcup_{i \in N \setminus S} \{\{i\}\}$, then $v(S, P)$ is called the **noncooperative core lower bound** of S. If $P = \{S, N \setminus S\}$, then $v(S, P)$ is called the **cooperative core lower bound** of S. Ambec and Ehlers (2008) showed that the *noncooperative*

[15] Harris (1985) shows that extensive games of perfect information have a subgame-perfect equilibrium if payoff functions are continuous and action sets are compact and lower hemi-continuous in history. Action sets are allowed to have infinite elements, such as a closed interval.

[16] Theorem 2 (and the statement of the last paragraph in p. 624) of Harris (1985) imply that the set of histories on the path of subgame-perfect equilibria is compact. Therefore, $v(S, P)$ is well-defined.

[17] We thank an associate editor for his/her insightful comments on this point.

[18] Reaction function of agent i is a function from history to an element of $\arg\max_{f_i'} B_i^P(h_n((f_{-i}, f_i'), h))$. A function $f \colon X \to \mathbb{R}$ is upper semi-continuous if $\{x \in X \colon f(x) > a\}$ is open for any $a \in \mathbb{R}$.

core lower bound of each coalition is no greater than any core lower bound. However, such a relation no longer holds when the cost of E_i is incorporated. To assert this observation, we provide a counterexample in Section B. We introduce a new notion of lower bounds as follows: For each $S \subseteq N$, let

$$\underline{v}(S) \equiv \min_{P \in \mathcal{P}(S)} v(S, P) \tag{1}$$

be the **pessimistic core lower bound** of S. The minimizer of the right-hand side of eq. (1) is the worst possible coalition structure against S. $\underline{v}(S)$ guarantees at least S's coalitional worth attainable by the water the coalition controls. The *pessimistic core lower bound* of each coalition is no greater than any core lower bound, even if water inflow causes disutility. $z \in \mathbb{R}^n$ is an element of **the pessimistic core** if $\sum_{i=1}^{n} z_i = \underline{v}(N)$, and for each $S \subset N$, $\sum_{i \in S} z_i \geq \underline{v}(S)$.

The **aspiration upper bound** of each coalition is defined as follows: For each $S \subseteq N$, let

$$\bar{v}(S) \equiv \max_{(x_j)_{j \in U_S}} \sum_{i \in S} b_i(x_i, E_i)$$

under the constraint that for each $j \in U_S$, $x_j \in [0, E_j]$.

Ambec and Sprumont (2002) and Ambec and Ehlers (2008) showed a unique allocation satisfying *aspiration upper bounds* and *core lower bounds*, which is called the **Downstream Incremental Distribution** (**DI distribution** for short). The *DI distribution*, denoted as $z^* \in \mathbb{R}^n$, is given by

$$z_i^* = \max_{(x_j)_{j \in U_i}} \sum_{j \in U_i} b_j(x_j, E_j) - \max_{(x_j)_{j \in U_i^0}} \sum_{j \in U_i^0} b_j(x_j, E_j).$$

Because the maximized welfare of each coalition U_i is unique, z_i^* is uniquely determined.

Lemma 1. *For each $i \in N$, z_i^* is uniquely determined.*

4.2 Core lower-bound properties

This section investigates the conditions under which the existence of a welfare distribution satisfying the *aspiration upper bounds* and *core lower bounds*. First, we demonstrate that even under Assumptions 1 and 2, the DI distribution fails to satisfy the core lower-bound property in the presence of disutility caused by water inflow. Furthermore, in this example, the pessimistic core is empty. The following example illustrates that if an agent negatively affected by water inflow prefers to dispose of some water downstream, then the DI distribution violates *all* core lower bounds.

Example 3. Let $N = \{1, 2, 3\}$. Let $b_1(x_1, E_1) = -(x_1 - 5)^2 + 21$, $b_2(x_2, E_2) = -1000 \max\{E_2 - 1/2, 0\} + 100 - (x_2 - 1/2)^2$, and $b_3(x_3, E_3) = -(x_3 - 5)^2 + 25$. Let $e_1 = 4$, $e_2 = 1/2$, and $e_3 = 0$. The DI distribution, $z^* = (z_1^*, z_2^*, z_3^*)$, is calculated as follows:

$$\begin{aligned}
z_1^* &= 20 \\
z_1^* + z_2^* &= \max_{x_1 \in [0,e_1],\ x_2 \in [e_1+e_2-x_1]} \left[-(x_1 - 5)^2 + 21 - 1000(e_1 - x_1) - (x_2 - 1/2)^2 + 100\right] \\
&= 120 \\
z_1^* + z_2^* + z_3^* &= \max_{x_1 \in [0,e_1],\ x_2 \in [e_1+e_2-x_1]} \left[\begin{array}{c} -(x_1 - 5)^2 + 21 - 1000(e_1 - x_1) - (x_2 - 1/2)^2 \\ + 100 - (e_1 + e_2 - x_1 - x_2 - 5)^2 + 25 \end{array}\right] \\
&= 124.5.
\end{aligned}$$

Therefore, $z^* = (20, 100, 4.5)$. The water consumption corresponding to z^*, denoted as x^*, is $x^* = (4, 0, 1/2)$.

Let $S = \{1, 3\}$ and $P = \{S, \{2\}\}$. Agent 2's optimal water consumption is $x_2^P = 1/2$. Given that S maximizes $\sum_{i \in S} b_i(x_i^P, E_i)$ in game (N, H, B^P), agent 1's optimal water consumption is x_1^P such that

$$x_1^P \in \arg\max_{x_1 \in [0,4]} \left[(x_1 - 5)^2 + 21 - (e_1 - x_1 - 5)^2 + 25\right] = 2.$$

The water consumption $x^P = (2, 1/2, 2)$ implies that $v(S, P) = b_1(x_1^P, E_1) + b_3(x_3^P, E_3) = 12 + 16 = 28 > 24.5 = \sum_{i \in S} z_i^*$. In this case, all core lower bounds coincide with each other because $P \setminus \{S\}$ $(= \{\{2\}\})$ takes the same optimal action in game (N, H, B^P).

Furthermore, in this example, the pessimistic core is empty. Suppose by contradiction that z is an allocation of the pessimistic core. By considering a deviation by agents 2 and 3, agent 1 consumes $x_1 = 4$, and agents 2 and 3 gain 104.5. Therefore, $z_2 + z_3 \geq 104.5$. Also, by considering a deviation by agents 1 and 2, $z_1 + z_2 \geq 120$. However, as shown above, it should be held that $z_1 + z_3 \geq 28$ so that z is a pessimistic core allocation. These inequalities imply $2(z_1 + z_2 + z_3) \geq 104.5 + 120 + 28 = 252.5$, which means $\sum_{i=1}^3 z_i \geq 126.25 > 124.5 = \sum_{i=1}^3 z_i^*$. This is a contradiction.

The key point of this example is that agent 2, which is suffering from water inflow, is incentivized to dispose of some water. This observation motivates the following condition:

Definition 1 (**Dam-up condition**). *The benefit function b_i satisfies **dam-up condition** if $x_i^*(E_i) = E_i$ for each $E_i > \hat{E}_i$.*

The dam-up condition implies that if agent i suffers from a flood, then it prefers to consume all its water inflow. This assumption can be interpreted

as agent i attempting to dam up the water inflow to prevent flooding. In this context, damming the water is understood as water consumption, as it reduces the inflow available to the downstream agents of i.

The following theorem demonstrates that the dam-up condition is sufficient for the existence of distribution rules that satisfy both the *aspiration upper bounds* and *core lower bounds*. That is, if the dam-up condition is satisfied, the DI distribution is the unique distribution satisfying both bounds. By contrast, if the dam-up condition is violated for some agent $i \in N$, there exists a profile of benefit functions of $N \setminus \{i\}$ such that the pessimistic core is empty. This statement concerns a natural generalization of benefit functions that appeared in Example 3.

Theorem 1. *(a) For any river-sharing problem, if b_i satisfies the dam-up condition for any $i \in N \setminus \{1, n\}$, then the DI distribution is the unique distribution that satisfies aspiration upper bounds and pessimistic core lower bounds.*

(b) If the benefit function $\tilde{b}_i$ violates the dam-up condition, then there exists a river-sharing problem (N, e, b), with $b_i = \tilde{b}_i$ for some $i \in N \setminus \{1, n\}$ such that the pessimistic core is empty.

Note that for constructing a river-sharing problem (N, e, b) for which the pessimistic core is empty, except for agent i, benefit functions are those of Ambec and Ehlers (2008). With the following benefit functions, theorem 1 (b) holds. These are quadratic functions and drought type; $b_{i-1}(x_{i-1}, E_{i-1}) = -\alpha(x_{i-1} - \beta)^2$ and $b_{i+1}(x_{i+1}, E_{i+1}) = -\alpha(x_{i+1} - \gamma)^2$, for some $\alpha, \beta, \gamma > 0$. These functions are similar to those in Example 3.

It is worth noting that the emptiness of the pessimistic core implies that the DI distribution violates the pessimistic core lower bounds, and therefore, this is a clear difference from Ambec and Ehlers (2008).

Another aspect of Example 3 is that the agent negatively affected by water inflow is located upstream of an agent that needs it. This insight leads us to consider scenarios where drought-type agents are positioned upstream of flood-type agents.

Let $D = \{i \in N \mid \hat{E}_i = \infty\}$ and $F = \{i \in N \mid \hat{E}_i \leq e_i\}$. D is the set of drought-type agents, and F is that of flood-type agents.

The following theorem shows that if the benefit functions violate the dam-up condition, then drought-type agents should be located upstream of flood-type agents for the DI distribution to satisfy the *aspiration upper bound* and *pessimistic core lower bound*. By contrast, if the location on the river does not satisfy such a condition, there exists a water problem for which the pessimistic core is empty.

Theorem 2. *(a) For any river-sharing problem, the DI distribution is the unique distribution that satisfies the aspiration upper bound and pessimistic core lower bound if $D = \{1, \ldots, k\}$ and $F = \{k+1, \ldots, n\}$ for some $k \in N$, even when the dam-up condition is violated.*

(b) Suppose that b^*_{i+1} is differentiable. Let $x^{U_i} = \arg\max_x \sum_{j \in U_i} b_j(x_j, E_j(x))$ s.t. $x_j \leq E_j(x)$. Consider a river-sharing problem satisfying

(i) $e_{i-1} > 0$ and

(ii) $i+1 \in D$ and $(b^*_{i+1})'(e_{i+1}) > \max\{\frac{\partial}{\partial x_{i-1}} b_{i-1}(x^{U_{i-1}}_{i-1}, E_{i-1}(x^{U_{i-1}})), 0\}$

for some $i \in N \setminus \{1, n\}$.[19] Then, there exists $\tilde{b}_i$ with $i \in F$ such that the pessimistic core for (N, e, b) with $b_i = \tilde{b}_i$ is empty.

Theorem 2 (b) assumes that a flood-type agent is located upstream of a drought-type agent, which has a significantly greater need for water than agent $i-1$. In the proof, $\tilde{b}_i(x_i, E_i) = -\alpha \cdot (x_i - \beta)^2 - \gamma \cdot (E_i + 1)^2$ for some $\alpha, \beta, \gamma > 0$. This assumption and the function $\tilde{b}_i$ capture the key property of the benefit functions in Example 3. Indeed, they satisfy conditions (i) and (ii). Under this assumption, the pessimistic core is empty, and therefore, the DI distribution fails to satisfy the pessimistic core lower bound property.

Remark 3. From Theorem 1, we immediately prove that when $n = 2$, the DI distribution satisfies both the aspiration upper bound and the pessimistic core lower bound without assuming that these agents' benefit functions satisfy the dam-up condition. This is because Theorem 1 (a) requires no restriction on the benefit functions of agents 1 and n. This result also implies that if only agents 1 and n are affected by flooding (i.e., the others are drought-type agents, or $\hat{E}_i = \infty$), then the DI distribution satisfies these bounds.

Furthermore, Theorem 2 indicates that if $N = F$, then the DI distribution meets both the aspiration upper bound and the pessimistic core lower bound.

5 Implementation

We introduce a simple strategic bargaining process to implement the DI distribution, called *upstream sequential bargaining*. Its protocol is as follows: Let $a(i) = n + 1 - i$ denote the proposer at stage $i \in N$. The bargaining process is as follows:

1. Let $R_0 = \emptyset$. In stage $i \in \{1, 2, \ldots, n-1\}$, agent $a(i)$ offers $(\tilde{E}_{a(i)}, \tilde{\rho}_{a(i)}) \in \mathbb{R}_+ \times \mathbb{R}$ to agent $a(i+1)$ that is $a(i)$'s immediate upstream agent.

[19] For the benefit functions of agents $N \setminus \{i\}$, whether the dam-up condition is satisfied is inconsequential.

2. Agent $a(i+1)$ decides whether to accept or reject the offer. If agent $a(i+1)$ rejects the offer, then revise $R_i = \{j \in N\colon\ j \geq a(i)\}$. R_i is the set of agents whose offers become inactive at this time. Otherwise, $R_i = R_{i-1}$. In either case, the bargaining proceeds to stage $i+1$.

3. Let $R = R_{n-1}$. In stage n, for each $k \in N \setminus R$ such that $k+1 \in N \setminus R$, agent k consumes the amount of water $\tilde{x}_k \equiv \tilde{E}_k - \tilde{E}_{k+1} + e_{k+1}$ and makes a payment $\tilde{\rho}_k$ to agent $k-1$.[20] For $k \in N \setminus R$ such that $k+1 \in R$, agent k chooses water consumption to maximize its utility, subject to $x_k \leq \tilde{E}_k$.

   By contrast, for each $j \in R$, given the water consumption of agents in $N \setminus R$, $(x_k)_{k \in N \setminus R} = (\tilde{x}_k)_{k \in N \setminus R}$, agent j chooses its consumption x_j to maximize its own benefit and pays nothing. That is, for each $j \in R$, $x_j = x_j^*(e_j + \tilde{E}_{j-1}(x) - x_{j-1})$.[21] and $\tilde{\rho}_j = 0$.

In this bargaining process, agent $a(i+1)$ may reject the proposal of its immediate downstream agent $a(i)$. This is because $a(i+1)$ will not have to flow to $a(i)$, the amount of water the other party offers if $a(i+1)$ does not want it.

Theorem 3. *In the upstream sequential bargaining, the DI distribution is the unique subgame-perfect equilibrium outcome.*

Remark 4. The upstream sequential bargaining process assumes that each rejected agent behaves noncooperatively. We can modify the model so that rejected agents can negotiate among themselves.

Let i^* be the last rejected agent in the upstream sequential bargaining process, i.e., the agent with the smallest index among all rejected agents. Then, let the endowment of agent i^* be $\tilde{E}_{i^*} = (\arg\max_{x \leq \tilde{E}_{i^*-1}} b_{i^*-1}(x, \tilde{E}_{i^*-1})) + e_{i^*}$. The rejected agents restart the upstream sequential bargaining process, from stage 1 to $n - i^* - 1$, among themselves. Even in this modified process, the payoff gained by agent i^* in the renegotiation is at most $\max_{x \leq \tilde{E}_{i^*}} b_{i^*}(x, \tilde{E}_{i^*})$ if the renegotiating agents all follow the strategy profile specified in the proof of Theorem 3. Therefore, the equilibrium strategy in the original bargaining process remains on the equilibrium path in the modified process.

Note that the bargaining process in the reverse order, or **downstream** sequential bargaining, does not implement the DI distribution.[22]

[20] The total monetary transfer for agent k is $m_k = \tilde{\rho}_{k+1} - \tilde{\rho}_k$.

[21] Since R is written as $\{k, k+1, \ldots, n\}$, $E_{j-1}(x)$ and x_{j-1} are well-defined before x_j is defined.

[22] Indeed, we have a counterexample. Let $N = \{1, 2\}$, $b_1(x_1) = -(x_1 - 5)^2 + 25$, $b_2(x_2) = -(x_2 - 5)^2 + 25$, and $(e_1, e_2) = (5, 3)$. In this case, $(x_1, x_2) = (4, 4)$ maximizes the total benefits. Then, the DI distribution is $(z_1^*, z_2^*) = (25, 23)$. In this case, the monetary

6 Discussion

To our knowledge, our model is the first attempt at modeling a river-sharing problem that considers the disutility of water inflow. Using this model, the present study contributes to analyses of the role of international environmental law and international negotiations in resolving conflicts over international rivers. International environmental law recommends that the International River Commission handles water use in compliance with such doctrines as ATS and UTI. The present study establishes a cooperative/noncooperative theoretical foundation for this recommendation via international environmental law. Moreover, in situations where soft laws, such as international environmental law, are not legally binding for the management of international rivers, we highlight the importance of an appropriate negotiation process for attaining fair international agreements.

Our model relates to analyses of simple hierarchical structures represented by line graphs.

Previous studies have analyzed how assigning authority and constraints appropriately within a hierarchy enables stable, efficient coordination (Demange 2004) and addressed fairness and incentives within the hierarchy regarding the division of joint revenues among entities forming the hierarchical structure (Hougaard et al. 2015). The simplest hierarchical structure models are those represented by line graphs. Linear hierarchical models can be applied to the economic analysis of the supply chain of technological knowledge,[23] and they are relevant to our model of the river-sharing problem.

The technological knowledge possessed by each firm in the supply chain corresponds to the water source of each country in the river-sharing model. Each firm's technological knowledge is assumed to be protected by intellectual property rights, but firms are assumed to be able to disclose this knowledge voluntarily to be utilized by downstream firms in the supply chain. Each firm engages in production activities using its knowledge and knowledge flowing from upstream. However, an excessive inflow of new knowledge may reduce a firm's profit. It can create conflicts with existing production processes and cause confusion among employees. Even if the firm decides not to utilize the inflowing knowledge, additional resources — such as managerial effort and

allocation is $(m_1, m_2) = (1, -1)$.

In downstream sequential bargaining, agent 1 makes a take-it-or-leave-it offer to agent 2, with $(E_1 - x_1, \tilde{\rho}_2) = (1, 3)$. Here, $\tilde{\rho}_2$ is agent 2's payment for agent 1. Then, agent 1's payoff is 27, and agent 2's payoff is 21, which differs from the DI distribution.

[23]Industries have many examples related to the supply chain of technological knowledge. For instance, the supply chain in semiconductor processing spans from upstream to downstream, involving foundational material manufacturers, equipment manufacturers, high-performance-chip producers, and device manufacturers that integrate semiconductors into end products.

personnel — may be required for its management.

This supply chain model is almost identical to the proposed river-sharing model. Regarding this technological knowledge supply chain, our findings suggest a way to enable firms within the supply chain to achieve efficient and stable coordination.

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A Omitted Proofs

Proof of Lemma 1. We show that the following problem has a solution for each $j \in N$.

$$\max_{(x_i,E_i)_{i\in U_j}} \sum_{i\in U_j} b_i(x_i, E_i) \text{ s.t., } \begin{cases} x_i \in [0, E_i], \\ E_{i+1} = E_i - x_i + e_{i+1}, \\ E_1 = e_1. \end{cases}$$

Let $\mathcal{C}_j = \{(x_i, E_i)_{i\in U_j} : x_i \in [0, E_i], E_{i+1} = E_i - x_i + e_{i+1}, \text{ and } E_1 = e_1\}$. Note that $\mathcal{C}_j$ is non-empty, compact, and convex as the constraint is a system of linear equations. Then, the solution exists, and the value does not depend on $(x_j)_{j\notin U_i}$. Therefore, the maximum value is uniquely determined. By the definition, z^* is also uniquely determined. □

Proof of Theorem 1. **Proof of (a):** Let z^* be the DI distribution.[24] The uniqueness immediately follows from the fact that if welfare distribution z

[24]This proof is a modification of Ambec and Ehlers (2008) Theorem 1.

satisfies aspiration upper bound and core lower bound, for each $i \in \{1, \dots, n\}$

$$\max_{(x_j)_{j \in U_i}} \sum_{j \in U_i} b(x_j, E_j(x)) \geq \sum_{j \in U_i} z_j \geq \max_{(x_j)_{j \in U_i}} \sum_{j \in U_i} b(x_j, E_j(x)).$$

This implies that $z_i = z_i^*$ for each $i \in \{1, \dots, n\}$.

Next, we show that z^* satisfies both bounds. Consider a coalition S. As for the aspiration upper bounds, as the water consumption that achieves the DI distribution satisfies the constraints $x_i \in [0, E_i]$, each value of DI distribution is no greater than $\bar{v}(S) = \max_{(x_j)_{j \in U_S}} b_i(x_i, E_i)$. Then, the DI distribution satisfies the aspiration upper bounds.

Now, we will show that the DI distribution satisfies the pessimistic core lower bound.

Step 1. First, we assume that S is a consecutive set. That is, $S = \{i \in N \colon \ell_0 \leq i \leq \ell_1\}$ for some $1 \leq \ell_0 < \ell_1 \leq n$. Consider the following partition:

$$P = \{\{1, \dots, \ell_0 - 1\}, S, \{\ell_1 + 1, \dots, n\}\}.$$

Then, as $v(S, P) = \max_{y_i} \sum_{i \in S} b_i(y_i, E_i(y))$,

$$v(S, P) + \max_y \sum_{i \in U^0_{\ell_0}} b_i(y_i, E_i(y)) \leq \max_x \sum_{i \in U_{\ell_1}} b_i(x_i, E_i(x)),$$

which implies that

$$v(S, P) \leq \max_x \sum_{i \in U_{\ell_1}} b_i(y_i, E_i(y)) - \max_y \sum_{i \in U^0_{\ell_0}} b_i(y_i, E_i(y)) = \sum_{i \in S} z_i^*.$$

Step 2. Consider an arbitrary coalition S and let $P = \{S\} \cup \bigcup_{i \in N \setminus S} \{\{i\}\}$. Let $x^P = h_n(f, \varnothing)$ for $f \in SPE(\mathcal{G}^P)$. As $e_i = x_i^*(e_i)$ for all $i \in N$, $E_{\min S}(x^P) = e_{\min S}$. Let $\{S_1, \dots, S_L\}$ be the coarsest consecutive partition of S with $S_1 < \dots < S_L$.[25] Choose the minimal l such that $E_{\max S_l + 1}(x^P) = e_{\max S_l + 1}$, and let $T_1 = \bigcup_{i=1}^{l} S_i$. This implies that $E_{\min S_{l+1}}(x^P) = e_{\min S_{l+1}}$ as each agent $k \in N \setminus S$ with $\max S_l + 1 \leq k \leq \min S_{l+1} - 1$ consumes its own endowments.

Next, choose the minimal $l' > l$ such that $E_{\max S_{l'} + 1}(x^P) = e_{\max S_{l'} + 1}$, and $T_2 = \bigcup_{i=l+1}^{l'} S_i$. Continuing this process, we constitute a partition $\mathcal{T} = \{T_1, \dots, T_M\}$ of S. By construction, $T_1 < \dots < T_M$.

For each $T \in \mathcal{T}$, let $c(T)$ be the largest consecutive set that contains T. Formally, $c(T) = \{i \in N : \min T \leq i \leq \max T\}$. Also, by construction, we

[25] Formally, each S_i is a consequitive set, $\bigcup_{i=1}^{L} S_i = S$, and $S_i \cap S_j = \emptyset$ for each $i, j \in \{1, \dots, L\}$.

have the following claim

Claim 1. *For each $i \in c(T) \setminus T$, (i) $i \in \{2, \dots, n-1\}$, (ii) $E_i(x^P) \leq \hat{E}_i$, and (iii) $x_i^P = x_i^*(E_i(x^P))$.*

Proof of Claim 1. Statement (i) immediately follows from the fact that $i \in c(T) \setminus T$.

The proof for statements (ii) and (iii) is as follows. Suppose that $E_i(x^P) > \hat{E}_i$. Then, by the dam-up condition, $x_i^P(E_i(x^P)) = E_i(x^P)$. Similarly, if (iii) fails, it also holds that $x_i^P(E_i(x^P)) = E_i(x^P)$. Therefore, if (ii) or (iii) is violated, such an agent i exists. Consider the smallest i such that $x_i^P(E_i(x^P)) = E_i(x^P)$.

For such i, let $S_j \in \{S_1, \dots, S_L\}$ such that (a) $S_j \subset T$ and (b) $\max S_j < i < \min S_{j+1}$. We will show that agents in S_j have no incentive to release water for S_{j+1}. To see this formally, consider agent $j^* = \max S_j$. If $x_{j^*}^*(E_{j^*}(x^P)) = E_{j^*}(x^P)$, as passing water has no effect on the utility of agents in $\bigcup_{\ell \geq j+1} S_\ell$, $x_{j^*}^P(E_{j^*}(x^P)) = x_{j^*}^*(E_{j^*}(x^P)) = E_{j^*}(x^P)$. This is a contradiction. Therefore, $x_{j^*}^*(E_{j^*}(x^P)) < E_{j^*}(x^P)$. By the assumption that $x_{j^*}^*(e_{j^*}) = e_{j^*}$, this implies that $e_{j^*} < E_{j^*}(x^P)$, and therefore, agent $j^* - 1$ passes the water. Applying the same discussion, $x_{j^*-1}^*(E_{j^*-1}(x^P)) < E_{j^*-1}(x^P)$. Using the same logic for each agent $\ell \in S_j$, we can show that $x_\ell^*(E_\ell(x^P)) < E_\ell(x^P)$. This implies that agents in S_j receive unnecessary water. Since each agent $\ell \notin S$ maximizes its own utility, $x_\ell^*(E_\ell(x^P)) < E_\ell(x^P)$ and $e_\ell < E_\ell(x^P)$. Then, increasing the amount of water released by agent $\max S_{j-1}$ increases the water inflow of agents in S_j. As this never increases the welfare of S_j, $x_{\max S_{j-1}}^*(E_{\max S_{j-1}}(x^P)) < E_{\max S_{j-1}}(x^P)$ and $e_{\max S_{j-1}} < E_{\max S_{j-1}}(x^P)$. Using the same discussion, we can show that for any $\ell \in S_k$ such that $k \leq j$, $e_\ell < E_\ell(x^P)$. This is a contradiction.

Therefore, $E_{\max S_j + 1}(x^P) = e_{\max S_j + 1}$. This contradicts that $i \in c(T)$. $\square$

By abusing the notation, we write that $v(T, P) = \sum_{j \in T} b_j(x_j^P, E_j(x^P))$ for each $T \in \mathcal{T}$. As $P = \{S\} \cup \bigcup_{i \in N \setminus S} \{\{i\}\}$, we have

$$\begin{aligned}
\sum_{T \in \mathcal{T}} \sum_{i \in c(T)} z_i^* &\geq \sum_{T \in \mathcal{T}} \left[v(T, P) + \sum_{j \in c(T) \setminus T} v(\{j\}, P) \right] \\
&\geq \sum_{T \in \mathcal{T}} \left(v(T, P) + \sum_{j \in c(T) \setminus T} b_j(\bar{x}_j, E_j(x^P)) \right) \\
&= v(S, P) + \sum_{T \in \mathcal{T}} \sum_{j \in c(T) \setminus T} b_j(\bar{x}_j, E_j(x^P))
\end{aligned}$$

The first inequality comes from the fact that $c(T)$ is a consecutive set and step 1. The second inequality comes from the fact that the utility of agent

$j \in c(T) \setminus T$ is maximized as they consume water at its satiation point, and the water inflow causes no disutility at $E_j(x^P)$ as Claim 1 (ii) states that $E_j(x^P) \leq \hat{E}_j$.

Recall that by Assumption 1 (3), $b_j(\bar{x}_j, E_j) \geq b_j^*(E'_j)$ for any $E'_j \geq \hat{E}_j$. Therefore, $b_j(\bar{x}_j, E_j(x^P)) \geq z_j^*$ for each $j \in c(T) \setminus T$, since z_j^* represents the marginal gain from cooperation, which cannot exceed the benefit an agent can secure individually under favorable conditions (i.e., consuming at its satiation point with no flood damage).

Using this inequality, we have that

$$\begin{aligned}
\sum_{i \in S} z_i^* &= \sum_{T \in \mathcal{T}} \sum_{i \in T} z_i^* = \sum_{T \in \mathcal{T}} \sum_{i \in c(T)} z_i^* - \sum_{T \in \mathcal{T}} \sum_{j \in c(T) \setminus T} z_j^* \\
&\geq v(S, P) + \sum_{T \in \mathcal{T}} \sum_{j \in c(T) \setminus T} (b_j(\bar{x}_j, E_j(x^P)) - z_j^*) \\
&\geq v(S, P).
\end{aligned}$$

This implies that the DI distribution satisfies the pessimistic core lower bound.

Proof of (b): The rough sketch of the proof is as follows.

1. When the dam-up condition is violated, there is a water inflow $\tilde{E}_i$ such that (i) agent i dislikes more additional inflow, and (ii) agent i is willing to release water.

2. Then, constructing a water problem such that water release by agent $i-1$ improves the benefit of agent $i+1$.

3. Lastly, we check the deviation incentive of agents $i-1$ and $i+1$ (and other coalitions).

The following is the formal proof.

Suppose that $\tilde{b}_i$ violates the dam-up condition. We prepare the following claim.

Claim 2. *There is $(\tilde{E}_i, \tilde{y}) \in \mathbb{R}^2_+$ with $\tilde{E}_i \geq \hat{E}_i$ that satisfies the following properties.*

P1. $x_i^*(\tilde{E}_i + \epsilon) < x_i^*(\tilde{E}_i) + \zeta\epsilon$ *for small* $\epsilon > 0$.

P2. $b_i(x_i, \tilde{E}_i - \tilde{y}) = b_i(x_i, \tilde{E}_i - \tilde{y} + y)$ *for any* $y \in [0, \tilde{y}]$.

P3. $\tilde{E}_i - \tilde{y} + y - x_i^*(\tilde{E}_i - \tilde{y} + y) = y$ *for any* $y \in [0, \tilde{y}]$.

P4. $x_i^*(\tilde{E}_i) = \tilde{E}_i - \tilde{y}$.

Proof of Claim 2. As $\tilde{b}_i$ violates the dam-up condition, there is $E_i' \geq \hat{E}_i$ such that $x_i^*(E_i') < E_i'$. We have the following two cases, depending on whether a satiation point exists or not.

Case 1: Suppose that $x_i^*(\hat{E}_i) < \hat{E}_i$. This implies that the satiation point $\bar{x}_i < \hat{E}_i$. By Assumption 1, $b_i(x, \hat{E}_i) = b_i(x, E_i)$ for each $E_i \in [0, \hat{E}_i]$. This implies that the satiation point, $\bar{x}_i = \arg\max_x b_i(x, E_i)$ for each $E_i \in [0, \hat{E}_i]$ is well-defined.

Then, let $\tilde{E}_i = \hat{E}_i$ and $\tilde{y} = \tilde{E}_i - \bar{x}_i$. Therefore, with this definition, P2 holds. Also by the continuity of x_i^*, $x_i^*(E_i) = \bar{x}_i = \tilde{E}_i - \tilde{y}$ for each $E_i \in [\bar{x}_i, \hat{E}_i]$. This implies P3 and P4. Furthermore, by Assumption 2, $x_i^*(\tilde{E}_i + \epsilon) < x_i^*(\tilde{E}_i) + \zeta\epsilon$, which implies P1.

Case 2: Suppose that $x_i^*(\hat{E}_i) = \hat{E}_i$. In this case, by the continuity of x_i^*, there exists the largest value of E_i that satisfies $x_i^*(E_i) = E_i$ with $E_i \in [\hat{E}_i, E_i']$. In this case, let $\tilde{E}_i$ be the largest value. Also let $\tilde{y} = 0$. As $x_i^*(\tilde{E}_i + \epsilon) < \tilde{E}_i + \epsilon$ for any small $\epsilon > 0$, $x_i^*(\tilde{E}_i + \epsilon + \epsilon') < x_i^*(\tilde{E}_i + \epsilon) + \zeta\epsilon'$ for any $\epsilon' > 0$. By the continuity of x_i^*, $x_i^*(\tilde{E}_i + \epsilon') < x_i^*(\tilde{E}_i) + \zeta\epsilon'$, which shows P1.

The other properties, P2–P4, immediately follow from the definition of $\tilde{E}_i$ and $\tilde{y}$. □

Roughly speaking, $\tilde{E}_i$ is the least inflow such that agent i is willing to release water to agent $i+1$.

Using $\tilde{E}_i$ and $\tilde{y}$ in Claim 2, we construct a river-sharing problem (N, e, b) with $b_i = \tilde{b}_i$. Let $e_j = 0$ if $j \in N \setminus \{i-1, i\}$, $b_j(x_i, E_i) = 0$ for each $j \geq i+2$, and b_j is arbitrary function if $j \in \{1, \ldots, i-2\}$. Let $b_{i-1}(x_{i-1}, E_{i-1}) = -\alpha \cdot (x_i - \beta)^2$, and $b_{i+1}(x_{i+1}, E_{i+1}) = -\alpha \cdot (x_{i+1} - \gamma)^2$. Here, $\alpha, \beta > 0$ are constants. Let β satisfy inequality

$$((1-\zeta) \cdot (\tilde{E}_i + \beta) - \tilde{y})^2 \geq 2(2\beta + 2\tilde{E}_i + \tilde{y})\tilde{y}. \tag{2}$$

This is possible as $\zeta < 1$, and the left hand side is a quadratic function of β while the right hand side is linear in β. Now define endowments of agents $i-1$ and i, and $i+1$'s satiation point γ as follows.

$$\begin{aligned} e_{i-1} &= \beta \\ e_i &= \tilde{E}_i - \tilde{y} \\ \gamma &= \beta + \tilde{y} + \tilde{E}_i. \end{aligned} \tag{3}$$

Note that in both cases 1 and 2, $x_i^*(e_i) = e_i$, which satisfies our assumption that $x_j^*(e_j) = e_j$ for each $j \in N$. Therefore, (N, e, b) is a river-sharing problem.

Now we show that the pessimistic core is empty for the problem (N, e, b). The proof consists of two steps:

Step 1: *Characterizing the DI Distribution.* We first characterize the consumption profile and welfare allocation corresponding to the DI distribution. We find that at the optimum, there are $\delta_1, \delta_2 \geq 0$ that satisfy (1) the water inflows are $E_i = \tilde{y} + \delta_1 + e_i$, $E_{i+1} = \tilde{y} + \delta_2$ and $z^*_{i+1} \leq -\alpha \cdot (\beta + \tilde{E}_i - \delta_2)^2$, and (2) $\delta_1, \delta_2 \to 0$ as $\alpha \to 0$.

To this end, we calculate the DI distribution as follows.

$$\begin{aligned} z^*_{i-1} &= 0 \\ z^*_{i-1} + z^*_i &= \max_{\theta} -\alpha \cdot (-\theta)^2 + b_i(x^*_i(\theta + \tilde{E}_i - \tilde{y}), \theta + \tilde{E}_i - \tilde{y}) \\ z^*_{i-1} + z^*_i + z^*_{i+1} &= \max_{\theta_1, \theta_2} -\alpha \cdot (-\theta_1)^2 + b_i(\theta_1 + \tilde{E}_i - \theta_2 - \tilde{y}, \theta_1 + \tilde{E}_i - \tilde{y}) \\ &\quad -\alpha \cdot (\theta_2 - \gamma)^2. \end{aligned}$$

Here, θ denotes the water outflow of agent $i-1$ when considering the problem that maximizes the welfare of agents $i-1$ and i. When considering to maximize the welfare of agents $i-1$, i, and $i+1$, θ_1 denotes the outflow of agent $i-1$ and θ_2 denotes that of agent i.

In each case, as the agent i needs no more water amount than $\tilde{E}_i - \tilde{y}$, $\theta = 0$ is optimal for $i-1$ and i. This shows that $z^*_i = b_i(x_i(\tilde{E}_i - \tilde{y}), \tilde{E}_i - \tilde{y})$.

Next, we consider the optimal values of θ_1 and θ_2. When $\alpha = 0$, $x^*_i(\theta_1 + \tilde{E}_i - \tilde{y})$ is optimal for agent i given θ_1. In other words, $\theta_2 = \theta_1 + \tilde{E}_i - y - x^*_i(\theta_1 + \tilde{E}_i - \tilde{y})$. Then, $b_i(\theta_1 + \tilde{E}_i - \tilde{y} - \theta_2, \theta_1 + \tilde{E}_i - \tilde{y}) = b^*_i(\theta_1 + \tilde{E}_i - \tilde{y})$. The value $b^*_i(\theta_1 + \tilde{E}_i - \tilde{y})$ is decreasing in θ_1 if $\theta_1 \geq \tilde{y}$ as it implies that $E_i \geq \hat{E}_i$. Then, $\theta_1 \leq \tilde{y}$ at the optimum. Further, in this case, by P3, $\theta_2 = \theta_1 + \tilde{E}_i - \tilde{y} - x^*_i(\theta_1 + \tilde{E}_i - \tilde{y}) = \theta_1$.

On the contrary, without consideration of agent i, specification eq. (3) implies that the marginal utility of water consumption is greater for $i+1$ than $i-1$ as long as $\theta_1 \leq \tilde{y}$. Indeed, while the marginal utility of agent $i-1$ with respect to water consumption is $2\alpha\theta_1$, that of agent $i+1$ is $2\alpha(\gamma - \theta_1) = 2\alpha(\beta + \tilde{y} + \tilde{E}_i - \theta_2)$. Given θ_1, $\theta_2 \leq e_i + \theta_i = \tilde{E}_i - \tilde{y} + \theta_i$. Then, the marginal utility is no less than $2\alpha(\beta + 2\tilde{y} - \theta_1)$, which is strictly greater than $2\alpha\theta_1$ if $\theta_1 \leq \tilde{y}$.

Note also that the agent i's utility is the same for any $\theta_1 \in [0, \tilde{y}]$ by P2. This implies that $\theta_1 = \tilde{y}$ is optimal.

Suppose that α is a small positive real number. Note that agent $i+1$'s marginal utility of consuming water is greater than that of agents $i-1$ and i in a neighborhood of $\theta_1 = \theta_2 = \tilde{y}$. Then, to maximize $\sum_{j \leq i+1} b_j$, $\theta_1 = \tilde{y} + \delta_1$ and $\theta_2 = \tilde{y} + \delta_2$ for some small $\delta_1, \delta_2 \geq 0$. These values depend on α, and are 0 if $\alpha = 0$. Then, $z^*_{i+1} \leq -\alpha \cdot (\beta + \tilde{E}_i - \delta_2)^2$.

Step 2: *The pessimistic core is empty.*

Suppose by contradiction that the pessimistic core is non-empty. Our strategy is to show that the sum of core constraints for coalitions $\{i-1, i\}$, $\{i, i+1\}$, and $\{i-1, i+1\}$ leads to a contradiction with the total available welfare.

Let $z \in \mathbb{R}^n$ be an element of the pessimistic core. Recall that $z^* \in \mathbb{R}^n$ is the DI distribution.

Note that $\sum_{j=1}^{k} z_j \geq \sum_{j=1}^{k} z_j^*$ for each k. If else, a coalition $\{1, \ldots, k\}$ can profitably deviate from the grand coalition. As agents in $\{1, \ldots, i-2\}$ have no water to transfer, we can focus on agents in $\{i-1, i, i+1\}$. Then, $\sum_{j=i-1}^{k} z_j \geq \sum_{j=i-1}^{k} z_j^*$ for each $k \geq i-1$.

By the assumption that $b_j(x_i, E_i) = 0$ for each $j \notin \{i-1, i, i+1\}$, $z_j^* = 0$. Therefore,

$$z_{i-1} + z_i + z_{i+1} = z_{i-1}^* + z_i^* + z_{i+1}^*.$$

We now examine core constraints for coalitions $\{i-1, i\}$, $\{i, i+1\}$, and $\{i-1, i+1\}$. Specifically, we calculate the difference $z_j + z_k - (z_j^* + z_k^*)$ for each $j, k \in \{i-1, i, i+1\}$ with $j \neq k$

1. Consider coalition $\{i-1, i\}$: $z_{i-1} + z_i \geq z_{i-1}^* + z_i^*$ as coalition $\{i-1, i\}$ yields welfare $z_{i-1}^* + z_i^*$.

2. Consider coalition $\{i, i+1\}$: as $e_{i-1} = \beta$, when we consider a deviation of coalition $\{i, i+1\}$, $x_{i-1}^P = \beta$ as no other agent wants a water for any partition P includes $\{i, i+1\}$. Then, to block the coalitional deviation, the following inequality should be satisfied:

$$z_i + z_{i+1} \geq \max_{\theta_2} \{b_i(\tilde{E}_i - E_2 - \tilde{y}, \tilde{E}_i - \tilde{y}) - \alpha \cdot (\theta_2 - \gamma)^2\}.$$

Moreover,

$$z_i^* + z_{i+1}^* = -\alpha(\tilde{y} + \delta_1)^2 + b_i(\tilde{E}_i - \tilde{y} - \delta_2 + \delta_1, \tilde{E}_i + \delta_1) - \alpha \cdot (\tilde{y} + \delta_2 - \gamma)^2.$$

Therefore, as $\tilde{E}_i + \delta_1 > \max\{\hat{E}_i, \tilde{E}_i - \tilde{y}\}$,

$$\begin{aligned}
&\max_{\theta_2} \{b_i(\tilde{E}_i - \theta_2 - \tilde{y}, \tilde{E}_i - \tilde{y}) - \alpha \cdot (E_2 - \gamma)^2\} \\
&\quad \geq \max_{\theta_2} \{b_i(\tilde{E}_i - \theta_2 - \tilde{y}, \tilde{E}_i - \tilde{y})\} - \alpha\gamma^2 \\
&\quad \geq b_i(\tilde{E}_i - \tilde{y} - \delta_2 + \delta_1, \tilde{E}_i - \tilde{y})\} - \alpha\gamma^2 \\
&\quad > b_i(\tilde{E}_i - \tilde{y} - \delta_2 + \delta_1, \tilde{E}_i + \delta_1)\} - \alpha\gamma^2.
\end{aligned}$$

This implies that

$$z_i + z_{i+1} - (z_i^* + z_{i+1}^*) > \alpha \left[(\tilde{y} + \delta_1)^2 + (\tilde{y} + \delta_2 - \gamma)^2 - \gamma^2\right].$$

3. Consider coalition $\{i-1, i+1\}$: Now we have the following claim:

Claim 3. *Consider coalition $S = \{i-1, i+1\}$, and arbitrary partition P, such that $S \in P$. Given other agents' strategies, let $U(\theta)$ be coalition S's welfare when $\theta = e_{i-1} - x_{i-1}$. Then, for small $\epsilon > 0$,*

$$U(\tilde{y}+\epsilon) - (z^*_{i-1} + z^*_{i+1}) > \alpha\Bigg[2\left(-\epsilon^2 + \left[(1-\zeta)\cdot(\tilde{E}_i + \beta) - \tilde{y}\right]\cdot\epsilon\right) \\ -\tilde{y}^2 + \delta_2(2(\beta + \tilde{E}_i) - \delta_2)\Bigg].$$

Proof of Claim 3. Consider coalition $S = \{i-1, i+1\}$, and arbitrary partition P, such that $S \in P$. In this partition, agents $j \leq i-2$, $x^P_j = 0$ since $e_j = 0$ for each $j \leq i-2$. Furthermore, agent i's consumption is $x^P_i(E_i) = x^*_i(E_i)$ for each E_i. This is because the maximum amount of water outflow by agent i is $\beta + \tilde{E}_i$, which is less than the satiation point of agent $i+1$. Then, even if agent i leaves a portion of water unconsumed more than her optimal, agent $i+2$'s inflow is unchanged. Therefore, independent of the coalition in which agent i participates, agent i's consumption is $x^*_i(E_i)$.

Then, $E_{i+1} = (e_{i-1} - x_{i-1} + \tilde{E}_i - \tilde{y}) - x^*_i(e_{i-1} - x_{i-1} + \tilde{E}_i - \tilde{y})$. If the outflow by $i-1$ is $\theta = e_{i-1} - x_{i-1}$, coalition S's welfare, denoted by $U(\theta)$, is then written as

$$U(\theta) = -\alpha\cdot(\theta)^2 - \alpha\cdot\left(x^*_i(\theta + \tilde{E}_i - \tilde{y}) + \beta + 2\tilde{y} - \theta\right)^2.$$

By $\zeta \in (0,1)$, P1 and P4, we have that

$$U(\tilde{y}+\epsilon) - U(\tilde{y}) > 2\alpha\cdot\left(-\epsilon^2 + \left[(1-\zeta)\cdot(\tilde{E}_i + \beta) - \tilde{y}\right]\cdot\epsilon\right).$$

In contrast, at the DI distribution, the welfare of $\{i-1, i+1\}$ is

$$z^*_{i-1} + z^*_{i+1} \leq -\alpha\cdot(\beta + \tilde{E}_i - \delta_2)^2$$

Note that

$$-\alpha\cdot(\beta + \tilde{E}_i - \delta_2)^2 - U(\tilde{y}) = -\alpha\tilde{y}^2 + \alpha\delta_2(2(\beta + \tilde{E}_i) - \delta_2).$$

Then, $U(\tilde{y}+\epsilon) > z^*_{i-1} + z^*_{i+1}$ is calculated as in the statement. □

As z is an element of the pessimistic core, $z_{i-1} + z_{i+1} \geq U(\tilde{y}+\epsilon)$. There-

fore,

$$\begin{aligned} z_{i-1} + z_{i+1} - (z^*_{i-1} + z^*_{i+1}) &\geq U(\tilde{y} + \epsilon) - (z^*_{i-1} + z^*_{i+1}) \\ &\geq \alpha \Bigg[2 \left(-\epsilon^2 + \left[(1-\zeta) \cdot (\tilde{E}_i + \beta) - \tilde{y} \right] \cdot \epsilon \right) \\ &\quad -\tilde{y}^2 + \delta_2 (2(\beta + \tilde{E}_i) - \delta_2) \Bigg]. \end{aligned}$$

Now summing up $z_j + z_k - (z^*_j + z^*_k)$ for each $j, k \in \{i-1, i, i+1\}$ with $j \neq k$ yields that

$$\begin{aligned} 0 &= \sum_{j,k \in \{i-1,i,i+1\}, j \neq k} [z_j + z_k - (z^*_j + z^*_k)] \\ &> \alpha \left[(\tilde{y} + \delta_1)^2 + (\tilde{y} + \delta_2 - \gamma)^2 - \gamma^2 \right] \\ &\quad + \alpha \left[2 \left(-\epsilon^2 + \left[(1-\zeta) \cdot (\tilde{E}_i + \beta) - \tilde{y} \right] \cdot \epsilon \right) - \tilde{y}^2 + \delta_2 (2(\beta + \tilde{E}_i) - \delta_2) \right]. \end{aligned} \tag{4}$$

As shown in step 1, when $\alpha \to 0$, $\delta_1, \delta_2 \to 0$. Therefore, the right-hand side of eq. (4) is positive for a small $\alpha > 0$ if

$$2 \left(-\epsilon^2 + \left[(1-\zeta) \cdot (\tilde{E}_i + \beta) - \tilde{y} \right] \cdot \epsilon \right) + \tilde{y}^2 - 2\gamma\tilde{y} > 0. \tag{5}$$

Recall that $\gamma = \beta + \tilde{y} + \tilde{E}_i$. Then, as β satisfies eq. (2), we can find $\epsilon > 0$ to satisfy inequality eq. (5). This implies that $0 = \sum_{j,k \in \{i-1,i,i+1\}, j \neq k} [z_j + z_k - (z^*_j + z^*_k)] > 0$. This is a contradiction, and therefore, the pessimistic core is empty. □

Proof of Theorem 2. **Proof of (a):** The proof for the uniqueness is the same as the proof of Theorem 1 (a). We can also prove that DI distribution satisfies the aspiration upper bound.

Now consider the pessimistic core lower bound. Take an arbitrary coalition S, and let $P = \{S\} \cup \bigcup_{i \in N \setminus S} \{\{i\}\}$ and $x^P = h_n(f, \varnothing)$ for $f \in SPE(\mathcal{G}^P)$. Recall that $x^{U_i} = \arg\max_x \sum_{i \in U_i} b_i(x_i, E_i(x))$ s.t. $x_i \leq E_i(x)$. Let $x^P_i(E_i) = f_i(h)$ such that $E_i = E_i(h)$ and $f \in SPE(\mathcal{G}^P)$ be the on-path action of the continuation strategy of the subgame-perfect equilibrium of the subgame in which the reached amount of water is E_i.

Consider $S \cap D$. Note that $E_i - x^*_i(E_i)$ never decreases in E_i by Assumption 2. If $x^*_i(E_i) = E_i$, $E_i - x^*_i(E_i) = 0$. If $x^*_i(E_i) < E_i$, $E_i + \epsilon - x^*_i(E_i + \epsilon) > E_i - x^*_i(E_i) + (1-\zeta)\epsilon > E_i - x^*_i(E_i)$. By the continuity of x^*_i, there is no point of decrease.

Then, as $e_i \leq x_i^*(e_i)$ for each $i \in N$, $S \cap D$ has no incentive to release water for $S \cap F$. This implies that $v(S \cap D, P) = v(S \cap D, P')$, where $P' = \{S \cap D\} \cup \{S \cap F\} \cup \bigcup_{i \in N \setminus S} \{\{i\}\}$. If $S \subset D$ as it is the subcase of Ambec and Ehlers (2008), $\sum_{i \in S} z_i^* \geq v(S, P)$. Therefore, $\sum_{i \in S \cap D} z_i^* \geq v(S \cap D, P)$. This also implies that $E_{k+1}(x^P) = e_{k+1}$.

Next, consider $S \cap F$. Note that all players in this set prefer less water inflow.

Recall that $P = \{S\} \cup \bigcup_{i \in N \setminus S} \{\{i\}\}$. Again, each $i \in S \cap F$ has no incentive to release water for its downstream $(S \cap F) \setminus \{1, \ldots, i\}$ as $x_i^*(e_i) = e_i$ for each $i \in N$. This also implies that $x_i^P = e_i$ for each $i \in S \cap F$.

Furthermore, note that $x_i = x_i^*(e_i) = e_i$ maximizes the overall utility as all agents prefer less water inflow. This proves that $v(S \cap F, P) = \sum_{i \in S \cap F} z_i^*$, and then, we conclude the proof.

Proof of (b): For some positive real numbers α, β, γ, let $b_i(x_i, E_i) = -\alpha \cdot (x_i - \beta)^2 - \gamma \cdot (E_i + 1)^2$ and $\beta = e_i$,[26] and let γ satisfy

$$\sup_{\theta \geq 0} \left[-\frac{\partial}{\partial x_{i-1}} b_{i-1}(x_{i-1}^{U_{i-1}} - \theta, E_{i-1}(x^{U_{i-1}})) + \sum_{j=i+1}^{n} \max\{(b_j^*)'(\theta + e_j), 0\} \right] < \gamma.$$

Such γ exists since the differentiability implies $\left|\frac{\partial b_j}{\partial x_j}\right| + \left|\frac{\partial b_j}{\partial E_j}\right| < \infty$ at any $x_j, E_j < \infty$ for each j.

With this benefit function, we calculate the DI distribution as follows.

$$\begin{aligned} z_{i-1}^* &= b_{i-1}^*(E_{i-1}(x^{U_{i-1}})) + C \\ z_{i-1}^* + z_i^* &= b_{i-1}(E_{i-1}(x^{U_{i-1}}), E_{i-1}(x^{U_{i-1}})) + b_i(\beta, \beta) + C \\ z_{i-1}^* + z_i^* + z_{i+1}^* &= b_{i-1}(E_{i-1}(x^{U_{i-1}})), E_{i-1}(x^{U_{i-1}})) + b_i(\beta, \beta) \\ &\quad + b_{i+1}(x_{i+1}^*(e_{i+1}), e_{i+1}) + C, \end{aligned}$$

where $C = \sum_{j=1}^{i-2} [b_j(x_j^{U_{i-1}}, E_j(x^{U_{i-1}})) - z_j^*]$.

We can further show that $x_{i-1}^{U_j}(E_{i-1}(x^{U_j})) = E_{i-1}(x^{U_j})$ for each $j \geq i+2$ as agent i's marginal disutility of water inflow is higher than other agents' marginal utility of water inflow.

We first show that the DI distribution violates the pessimistic core lower bound. Consider deviation by a coalition $S = N \setminus \{i\}$. Then, the unique possible partition is $P = \{S, \{i\}\}$. Then, $x_i^P = \beta = e_i$, and therefore, $E_i(x^P) - x_i^P = E_{i-1}(x^P) - x_{i-1}^P$.

Consider the consumption of agent $i-1$. We have two cases.

[26] This specification implies that $x_i^*(e_i) = e_i$.

Case 1. Consider the case $x^*_{i-1}(E_{i-1}(x^{U_{i-1}})) < E_{i-1}(x^{U_{i-1}})$. Then, let $x_{i-1} = x^*_{i-1}(E_{i-1}(x^{U_{i-1}}))$, and $x_j = x_j^{U_{i-1}}$ for each $j \leq i-2$. This profile improves agent $i-1$'s utility from the DI distribution. On the one hand, as agent i releases a positive amount of water to $i+1$ as $E_i > e_i$, agent $i+1$'s utility is strictly improved from the DI distribution. On the other hand, the welfare of the other agents in S (i.e., $\{1, \ldots, i-2, i+2, \ldots, n\}$) is unchanged from the DI distribution.

Case 2. Consider the case that $x^*_{i-1}(E_{i-1}(x^{U_{i-1}})) = E_{i-1}(x^{U_{i-1}})$. Consider the water consumption $x_{i-1} = x^*_{i-1}(E_{i-1}(x^{U_{i-1}})) - \epsilon$ for a small $\epsilon > 0$. Then, the amount of water reached by agent i is $E_i = \epsilon + e_i$. When $\epsilon > 0$ is small enough, $\epsilon < x^*_{i+1} - e_{i+1}$. In this case, $x_i^P(E_i) = e_i$ and $E_{i+1} = \epsilon + e_{i+1}$. Then, the sum of the utilities of agent $i-1$ and $i+1$ improves from the DI distribution by the assumption (ii) of the theorem. Here, the utilities of the other agents $S \setminus \{i-1, i+1\}$ are unchanged from the DI distribution.

In each case, as the welfare of coalition S can be strictly improved from the DI distribution, the DI distribution violates any core lower bounds.

Suppose by contradiction that the pessimistic core is non-empty. Let z be an element of the pessimistic core. First note that $\sum_{j=1}^k z_j \geq \sum_{j=1}^k z_j^*$ for each k. Otherwise, coalition U_k can deviate. Then, as $\sum_{j \in N} z_j = \sum_{j \in N} z_j^*$, $\sum_{j=k}^n z_j \leq \sum_{j=k}^n z_j^*$ for each k.

Consider a coalition deviation by $S = N \setminus \{i\}$. As shown that z^* violates the pessimistic core lower bounds, $\sum_{j \in S} z_j > \sum_{j \in S} z_j^*$.

Next, consider a coalition $\{i, \ldots, n\}$. Also consider an arbitrary partition P includes $\{i, \ldots, n\}$. We now show that $x_{i-1}^P = E_{i-1}(x^P)$. Suppose by contradiction that $x_{i-1}^P < E_{i-1}(x^P)$. Then, by the assumption that $x_j^*(e_j) = e_j$ for all j, $E_{i-1}(x^P) > e_{i-1}$. Then, this implies that $E_{i-2}(x^P) > e_{i-2}$, and $x^*_{i-2}(E_{i-2}(x^P)) < E_{i-2}(x^P)$. Otherwise, $i-2$ has no incentive to pass the water as it never increases the benefit of agent i. Continuing this process, we conclude that $E_j(x^P) > e_j$ for each j, which contradicts the assumption.

Note that as the marginal disutility of water inflow of agent i, γ is large enough, at the optimal consumption profile that maximizes the total benefit of N, $x_{i-1}^N = E_{i-1}(x^N)$. By, $x_{i-1}^P = E_{i-1}(x^P)$, the welfare of coalition $\{i, \ldots, n\}$ coincide with the welfare that of the maximized welfare. Then, $\sum_{j=i}^n z_j \geq \sum_{j=i}^n z_j^*$.

Therefore, we have shown that $\sum_{j=1}^i z_j \geq \sum_{j=1}^i z_j^*$, $\sum_{j=i}^n z_j \geq \sum_{j=i}^n z_j^*$, and $\sum_{j \neq i} z_j > \sum_{j \neq i} z_j^*$, which implies that $2\sum_{j=1}^n z_j > 2\sum_{j=1}^n z_j^*$. This is a contradiction, and thus the pessimistic core is empty. $\square$

Proof of Theorem 3. Define $\omega_t = (\tilde{E}_{N\setminus U_{a(t)}}, \tilde{\rho}_{N\setminus U_{a(t)}}, R)$, which specifies the past and active offers.

For each consecutive $S \subseteq N$ such that $1 \in S$, let $X^*(S) \in \mathbb{R}^n_+$ be such that

$$X^*_j(S) = \begin{cases} x_j(S) & \text{if } j \in S, \\ \arg\max_{x_j \le \sum_{k\le j} e_k - \sum_{k \le j-1} X^*_k(S)} b_j(x_j, E_j) & \text{otherwise.} \end{cases}$$

Clearly, $X^*(N \setminus R)$ is the final consumption bundle in stage n when the consumption level implied by the offers is $\tilde{x}_{N\setminus R} = x(N\setminus R)$. That is if $i \in N\setminus R$, this implies that each agent j ($j \le i$)'s offer is not rejected, in which case, by the definition of the bargaining process, agent i consumes $x_i(N\setminus R)$. Otherwise, some i's upstream agent's offer is rejected, and then, each agent $i \in R$ decides the consumption noncooperatively.

We also define[27]

$$B^*_{a(t)}(\omega_t) = \chi(a(t-1) \notin R)\tilde{\rho}_{a(t-1)} + \max_{x_{U_{a(t)}} \in Z(\omega_t)} \sum_{j \in U_{a(t)}} b_j(x_j, E_j),$$

where

$$Z(\omega_t) \equiv \left\{ x_{U_{a(t)}} \in \mathbb{R}^{U_{a(t)}}_+ : \begin{array}{ll} x_k \in [0, E_k] \text{ for each } k \le a(t), & \\ E_{a(t)} - x_{a(t)} = \tilde{E}_{a(t-1)} - e_{a(t-1)} & \text{if } a(t-1) \notin R \end{array} \right\}$$

Here, $x_k \le E_k$ is the physical constraint. By contrast, $E_{a(t)} - x_{a(t)} = \tilde{E}_{a(t-1)} - e_{a(t-1)}$ is the resource constraint to guarantee the demanded water inflow level $\tilde{E}_{a(t-1)}$ for agent $a(t-1)$. If $a(t-1)$'s offer is rejected, agent $a(t)$ need not meet this constraint.

Let $\tilde{x}^t(\omega_t) = \arg\max_{x_{U_{a(t)}} \in Z(\omega_t)} \sum_{j \in U_{a(t)}} b_j(x_j, E_j)$. Note that given a fixed R, for each t, t' with $t \le t'$, $\tilde{x}^t_i(\omega_t) = \tilde{x}^{t'}_i(\omega_{t'})$ for each $i \le a(t')$ when $\tilde{x}^{t'}_j = \tilde{x}^t_j(\omega_t)$ for each $j \in (a(t'), a(t)] \cap \mathbb{N}$. This is because payment $\tilde{\rho}$ does not affect the maximizer.

Using the notations above, we calculate the subgame-perfect equilibrium of the bargaining process by backward induction.

Step 1. Consider stage $n-1$. Let $\tilde{x}_{N\setminus U^0_2} = (\tilde{x}_j)_{j \in N\setminus U^0_2}$. We have three cases.

Case (1). Suppose that agent 2 accepted offer $(\tilde{x}_3, \tilde{\rho}_3)$ such that $B^*_2(\omega_{n-1}) \ge b_1(X^*_1(\emptyset), E_1) + b_2(X^*_2(\emptyset), E_2)$.

In this case, agent 2 offers so that $\tilde{x}_2 = \tilde{x}^{n-1}_2(\omega_{n-1})$ and $\tilde{\rho}_2 = z^*_1 - \pi_1(\tilde{x}^{n-1}_1(\omega_{n-1}))$. Then, agent 2's utility is $B^*_2(\omega_{n-1}) - b^*_1$ and agent 1's utility is b^*_1. As $z^*_1 = b_1(X^*_1(\emptyset), E_1)$, agents 1 and 2 prefer the offer to be accepted.

[27] χ is the indicator function: $\chi(E) = 1$ if E is true, and otherwise, 0.

Case (2). Suppose that agent 2 accepted offer $(\tilde{x}_3, \tilde{\rho}_3)$ such that $B_2^*(\omega_{n-1}) < b_1(X_1^*(\emptyset), E_1) + b_2(X_2^*(\emptyset), E_2)$.

As considered in case (1), agent 1 rejects the offer if she achieved utility is less than z_1^*. In contrast, if agent 1's achieved utility is no less than $z_1^* = b_1(X_1^*(\emptyset), E_1)$, agent 2's utility is less than $b_2(X_2^*(\emptyset), E_2)$. Then, agent 2 makes an offer that agent 1 rejects, in which case, agent 2 gains utility $b_2(X_2^*(\emptyset), E_2)$. Therefore, any offer is rejected.

Case (3). Suppose that agent 2 rejected offer $(\tilde{x}_3, \tilde{\rho}_3)$. In this case, as agents $i \geq 3$ decides the consumption level after decision of (x_1, x_2), and $m_3 = 0$, agent 2 offers $\tilde{x}_2$ and $\tilde{\rho}_1 = b_1^* - \pi_1(\tilde{x}_1)$. In this case, as $R = \{3, \ldots, n\}$, $B_2^*(\omega_{n-1}) = z_1^* + z_2^*$.

Step 2. Consider stage $t' \leq n-2$. We suppose an induction assumption: for each $t > t'$, in the continuation game, agent $a(t)$ offers $\tilde{E}_{a(t)}$ so that $\tilde{x}_{a(t)} = \tilde{x}_{a(t)}^t(\omega_t)$, and $\tilde{\rho}_{a(t)} = \sum_{j=1}^{a(t+1)}[z_j^* - b_j(\tilde{x}_{a(t)}^t(\omega_t), E_j)]$ if $B_{a(t)}^*(\omega_t) \geq b_{a(t)}(X_{a(t)}^*(U_{a(t+1)}), E_{a(t)}) + \sum_{j=1}^{a(t+1)} z_j^*$, and otherwise, agent $a(t)$ makes an offer whom agent $a(t+1)$ rejects. Agent $a(t+1)$ accepts agent $a(t)$'s offer if and only if $B_{a(t+1)}^*(\omega_{t+1}) \geq \sum_{j=1}^{a(t+1)} z_j^*$.

As already done for the case $t' = n-1$ in step 1, consider $t' < n-1$. First, we consider agent $a(t'+1)$'s decision. By the induction assumption, agent $a(t'+2)$ rejects the offer if and only if $B_{a(t'+2)}^*(\omega_{t'+2}) \geq \sum_{j=1}^{a(t'+2)} z_j^*$. Then, if agent $a(t'+1)$ accepts the offer, her payoff is $B_{a(t'+1)}^*(\omega_{t'+1}) - \sum_{j=1}^{a(t'+2)} z_j^*$. In contrast, if the agent rejects the offer, $B_{a(t'+1)}^*(\omega_{t'+1}) = \sum_{j \in U_{a(t'+1)}} z_j^*$. Therefore, the payoff is $z_{a(t'+1)}^*$. Then, we are done.

Consider agent $a(t')$'s decision. We have three cases.

Case (1). Suppose that $B_{a(t')}^*(\omega_{t'}) \geq b_{a(t')}(X_{a(t')}^*(U_{a(t'+1)}), E_{a(t')}) + \sum_{j=1}^{a(t'+1)} z_j^*$.

In this case, agent $a(t')$ offers as specified in the induction assumption. Then, agent $a(t)$'s utility is $B_{a(t')}^*(\omega_{n-1}) - \sum_{j=1}^{a(t'+1)} z_j^*$. By agent $a(t'+1)$'s decision, this is the maximal payoff under the premise that agent $a(t'+1)$ accepts the offer. In contrast, if the agent rejects it, $a(t)$'s utility is $b_{a(t')}(X_{s(t')}^*(U_{a(t'+1)}), E_{a(t')})$. Therefore, the specified offer is the best response.

Case (2). Suppose that $B_{a(t')}^*(\omega_{t'}) < b_{a(t')}(X_{a(t')}^*(U_{a(t'+1)}), E_{a(t')}) + \sum_{j=1}^{a(t'+1)} z_j^*$.

Then, as in case (2) for $t' = n-1$, agent $a(t'+1)$ rejects the offer.

Case (3). Suppose that $a(t')$ already rejected the offer by $a(t'-1)$. Then $B_{a(t)}^*(\omega_t) = \sum_{j=1}^{a(t')} z_j^*$. Note that $b_{a(t')}(X_{s(t')}^*(U_{a(t'+1)}), E_{a(t')}) \leq z_{a(t')}^*$. This is

because, if not, $\sum_{j\in U_{a(t'+1)}} b_j(x_j(U_{a(t'+1)}), E_j) + b_{a(t')}(X^*_{s(t')}(U_{a(t'+1)}), E_{a(t')}) > z^*_{U_{a(t')}} = \max_x \sum_{j\in U_{s(t')}} b_j(x_j, E_j)$. This is a contradiction. Therefore, $B^*_{a(t)}(\omega_t) \geq b_{a(t')}(X^*_{s(t')}(U_{a(t'+1)}), E_{a(t')}) + \sum_{j=1}^{a(t'+1)} z^*_j$. This case is already considered in case (1).

In each case, the induction statement is proven. Then, in this equilibrium, agent $a(t)$'s utility is

$$\begin{aligned}\tilde{\rho}_{a(t-1)} - \tilde{\rho}_{a(t)} + b_{a(t)}(\tilde{x}_{a(t)}(\omega_t), E_{a(t)}) &= z^*_{a(t)} - b_{a(t)}(\tilde{x}_{a(t)}(\omega_t), E_{a(t)}) + b_{a(t)}(\tilde{x}_{a(t)}(\omega_t), E_{a(t)}) \\ &= z^*_{a(t)},\end{aligned}$$

which concludes the proof. □

B Noncooperative and cooperative core lower bounds

Let $P_{NC} = \{S\} \cup \bigcup_{i\in N\setminus S}\{\{i\}\}$. The worth $v(S, P_{NC})$ is referred to as *noncooperative core lower bound*. Relatedly, $v(S, \{S, N \setminus S\})$ is referred to as *cooperative core lower bound*. Ambec and Ehlers (2008) showed that the noncooperative core lower bound of each $S \subseteq N$ is no greater than any core lower bound. In other words, the noncooperative core lower bound coincides with the pessimistic core lower bound (Ambec and Ehlers, 2008, Proposition 1 (2)). However, in our environment, if some agent suffers from water inflow, this may not hold.

Example 4. Let $N = \{1, 2, 3\}$. Let $b_1(x_1, E_1) = -(x_1 - 5)^2$, $b_2(x_2, E_2) = -(x_2 - 5)^2 - 100(E_2 - 4)^2$, and $b_3(x_3, E_3) = -(x_3 - 5)^2$. Let $e_1 = 4$, $e_2 = 5$, and $e_3 = 0$.

Let $S = \{2\}$, $P = \{\{1, 3\}, \{2\}\}$, and $P' = \{\{1\}, \{2\}, \{3\}\}$. We have that $v(S, P) = b_2(5, 7) = -900$, and $v(S, P') = b_2(5, 5) = -100$. Since $v(S, P') = -100 > -900 = v(S, P)$, the *noncooperative core lower bound* of S exceeds its *cooperative core lower bound*.

In this example, agent 2, who suffers from water inflow, violates the dam-up condition, as the following proposition shows.

Proposition 1. *When the dam-up condition holds, $v(S, P_C) \geq v(S, P_{NC})$.*

Proof. Let $\{S_1, \ldots, S_k\}$ be the coarsest consecutive partition of S, where each S_i is a consecutive set. In the coalition structure P_{NC}, as $e_i \leq x^*_i(e_i)$, $E_{\min S_1}(x^{P_{NC}}) = e_{\min S_1}$. Consider a subgame starting from the decision node of $i \in S_l$, and let agents in S_l leave water amount to agents in S_{l+1}. Then, agents between S_l

and S_{l+1} consume the optimal consumption level $x_i^*(E_i)$, and $E_i < \hat{E}_i$ for each $\max S_l < i < \min S_{l+1}$. This implies that they need water. Under coalition structure P_C, suppose that agents between S_l and S_{l+1} change the consumption bundle from P_{NC}. If it is profitable for them, they reduce the consumption of water, which increases the water inflow of agent $\min S_{l+1}$. This also increases the water inflow of agent $\max S_{l+1} + 1$. By the dam-up condition, for each $i \in S_{l+1}$, $E_i < \hat{E}_i$. This implies that the welfare of agents in S_{l+1} is improved for each subgame. □

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